

Symmetry-Resolved Tensor Topology for Computational Oversight of Wildlife Neural Networks: Spectral Diagonalization, Lie-Algebraic Structure, and Finite Convergence

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Abstract

Modern wildlife-monitoring systems increasingly depend on deep neural networks whose performance is conventionally assessed through accuracy, precision, recall, F1 score, mean average precision, computational cost, and model size. These measures, while essential, provide limited information about internal computational structure and offer weak guarantees concerning structural stability under transformation, compression, conversion, or deployment. This paper proposes a symmetry-resolved tensor-topological framework for computational oversight of neural wildlife models. The framework treats a neural network as a composition of tensor operators and constructs Hermitian Gram operators from model tensors, enabling spectral diagonalization and extraction of effective rank, spectral entropy, condition measures, and symmetry-sector weights. We introduce a Lie-algebraic representation of model transformations through generators satisfying $[T_a, T_b] = i f_{ab}^c T_c$, and decompose tensor representations into invariant or approximately invariant subspaces. The approach is motivated by symmetry-resolved exact diagonalization in quantum many-body systems and by Lie-group-equivariant neural networks. For topology, we construct filtration-dependent representations of neural tensors and use persistent homology to characterize structural features not captured by conventional parameter counts or FLOPs. We define a composite Symmetry-Resolved Tensor Complexity (STTC) and establish conditional monotonicity and finite-convergence results for iterative model-reduction procedures satisfying a discrete descent condition. The framework is proposed for analysis of ONNX representations, Microsoft MegaDetector, PyTorch-Wildlife, and MegaDetector-Acoustic models. The principal contribution is a computable oversight methodology for determining the degree to which symmetry, spectral organization, and topological stability are present; empirical values are intentionally not asserted until the released model artifacts are analyzed.

Keywords: neural network complexity; exact diagonalization; tensor topology; Lie algebra; symmetry; ONNX; MegaDetector; PyTorch-Wildlife; bioacoustics; wildlife AI; persistent homology; model oversight; spectral complexity.

1. Introduction

Deep neural networks have become an important component of automated biodiversity monitoring. Camera-trap systems generate large volumes of imagery, while passive acoustic monitoring produces long-duration recordings requiring automated detection and classification. Microsoft's MegaDetector is an open-source system for locating animals, people, and vehicles in camera-trap imagery, while PyTorch-Wildlife provides a broader framework containing detection models, species classifiers, and associated inference tooling.

More recently, MegaDetector-Acoustic extends this ecosystem to bioacoustic monitoring, providing models and tooling for audio-based wildlife detection and species classification, including Mel-spectrogram generation, ResNet-based classification, and sliding-window inference.

Conventional machine-learning metrics answer an important question: how accurately does the model perform the intended task? They do not completely answer what structural complexity of computation produces that performance. This distinction becomes important when a model is exported between frameworks, compressed, quantized, pruned, transformed into ONNX, deployed at the edge, or subjected to distribution shifts.

ONNX and related interchange representations provide an opportunity for structural auditing because the computational graph and tensor operators can be inspected independently of the original training environment. A second motivation comes from symmetry-resolved exact diagonalization in quantum many-body physics, where Hamiltonians are decomposed

into invariant sectors associated with symmetries. The same mathematical idea can be transferred cautiously to neural-network analysis: if a transformation group acts on tensors or representations, computational space can potentially be decomposed into invariant or approximately invariant sectors.

2. Problem Statement

Let a trained neural network be represented as $F_{\theta} = T_L \circ T_{(L-1)} \circ \dots \circ T_1$, where each T_i represents a parameterized tensor operator, nonlinear transformation, normalization, convolution, attention operation, or graph transformation.

Conventional complexity measures include $P(F)$ =number of parameters, $C_{\text{FLOP}}(F)$ =number of floating-point operations, and $A(F)$ =task performance. These quantities do not uniquely characterize internal structure. Two models may satisfy $A(F_1)$ approximately equal to $A(F_2)$ while possessing different spectral distributions, effective ranks, condition numbers, symmetry-breaking residuals, or topological organization.

We therefore seek a computable structural quantity $C(F)$ that can be calculated from model tensors or ONNX, is invariant or approximately invariant under specified transformations, exposes structure beyond parameter count and FLOPs, admits explicit bounds, supports a monotonicity criterion, and can be evaluated on real wildlife-monitoring models.

3. Relationship to Lie Computability

We use the term Lie computability operationally rather than claiming that a neural network is itself a physical realization of a particular simple Lie group. Let G be a transformation group acting on representation space H , with representation $\rho: G \rightarrow GL(H)$. For a continuous Lie group, infinitesimal generators T_a satisfy $[T_a, T_b] = i f_{ab}^c T_c$.

A neural representation is exactly equivariant when $F(\rho(g)x) = \rho'(g)F(x)$. The associated infinitesimal condition is $T'_a F = D F T_a$, where DF is the local derivative. We therefore measure the residual $R_a(F) = T'_a F - D F T_a$ and define an aggregate symmetry-breaking measure $R_{\text{sym}}(F) = \sum_a \|R_a(F)\|_F^2 / (\sum_a \|T'_a F\|_F^2 + \epsilon)$. Thus $R_{\text{sym}}=0$ corresponds to exact equivariance under the tested generators; increasing values indicate increasing symmetry breaking. The framework measures Lie-algebraic structure rather than assuming it.

4. Spectral Tensor Representation

For each model tensor W_i , define the Gram operator $H_i = W_i^\dagger W_i$. Since H_i is Hermitian positive semidefinite, the spectral theorem provides $H_i = U_i \Lambda_i U_i^\dagger$, with nonnegative eigenvalues. This is the central diagonalization operation.

The normalized spectral distribution is $p_{ij} = \lambda_{ij} / \sum_k \lambda_{ik}$. Define spectral entropy $S_i = -\sum_j p_{ij} \log p_{ij}$ and effective spectral rank $r_{\text{eff},i} = \exp(S_i)$. Effective rank distinguishes tensors with identical algebraic rank but very different concentration of spectral energy.

5. Symmetry-Resolved Tensor Decomposition

Suppose a finite or compact symmetry group has representation operators U_g . A symmetry sector is represented by projector P_λ . The tensor decomposes as $W = \sum_\mu (\lambda_\mu, \mu) P_\lambda W P_\mu$. Diagonal blocks represent symmetry-preserving computation, while off-diagonal blocks measure coupling between sectors.

Define symmetry leakage $L_{\text{sym}} = \sum_{\lambda \neq \mu} \|P_\lambda W P_\mu\|_F^2 / (\|W\|_F^2 + \epsilon)$. For exact symmetry, $L_{\text{sym}}=0$. The quantity can be interpreted as the fraction of Frobenius energy lying outside symmetry-preserving blocks.

6. Tensor Topology

Spectral analysis captures algebraic structure but not necessarily topology of learned representations. Let X_i be feature vectors generated by a model layer. Construct a metric $d(x_a, x_b)$ and filtration $K_\epsilon(X_i)$. Persistent homology computes $H_k(K_\epsilon)$ over filtration scales. The persistence diagram $D_k(X_i)$ contains birth-death pairs, and a scalar persistence functional can be defined as $T_k(X_i) = \sum_{(b,d) \in D_k} (d-b)^q$, $q \geq 1$.

This construction follows established topological-data-analysis methodology and is intended as a structural descriptor rather than a claim that every neural tensor has an intrinsic topological invariant independent of representation choices.

7. Symmetry-Resolved Tensor Complexity

We define the proposed complexity functional:

$$C_{STT}(F) = \sum_i \omega_i [\log(1+r_{eff,i}) + \beta L_{sym,i}] + \gamma \sum_i T_i$$

where $\omega_i \geq 0$, $\sum_i \omega_i = 1$, and $\beta, \gamma \geq 0$.

8. Proposition 1: Spectral Bound

Let W have r nonzero singular values and define p_i from squared singular values. Then $1 \leq r_{eff}(W) \leq r$. This follows because p is a probability distribution supported on at most r elements, so its Shannon entropy lies between 0 and $\log r$; exponentiation gives the result.

$$1 \leq r_{eff}(W) \leq \text{rank}(W)$$

9. Proposition 2: Symmetry Leakage Bound

With mutually orthogonal complete projectors P_λ , Frobenius orthogonality gives $\|W\|_F^2 = \sum_\lambda (\text{tr}(P_\lambda W P_\lambda))$. Therefore $0 \leq L_{sym} \leq 1$.

$$0 \leq L_{sym} \leq 1$$

10. Proposition 3: Monotonic Complexity Under Structure-Preserving Reduction

Consider $F_0, \dots, F_N$ generated by a reduction operator. If effective spectral rank, symmetry leakage, and topological persistence are each nonincreasing at every step, and all weights in the composite metric are nonnegative, then $C_{STT}(F_{k+1}) \leq C_{STT}(F_k)$. This follows because each constituent is assigned a nonnegative monotone contribution.

$$C_{STT}(F_{k+1}) \leq C_{STT}(F_k)$$

11. Proposition 4: Finite Convergence

Suppose the reduction procedure operates on a finite or discretized structural state space and every non-fixed-point reduction decreases complexity by at least $\delta > 0$. If C_{min} is the minimum attainable complexity, convergence occurs after at most $\text{ceil}((C_{STT}(F_0) - C_{min})/\delta)$ strict reductions. This is a conditional finite-convergence result for the defined reduction procedure; it does not imply convergence of arbitrary neural-network training.

$$N \leq \text{ceil}((C_{STT}(F_0) - C_{min})/\delta)$$

12. ONNX as an Auditing Representation

For each ONNX graph node $v_i: X_i \rightarrow X_{i+1}$, extract operator type, tensor dimensions, parameter tensors, singular spectrum, effective rank, symmetry representation, block leakage, and topological descriptors where applicable. The resulting graph becomes $G_{ONNX}=(V,E,S,T)$, where S contains spectral quantities and T contains topological quantities. A PyTorch-to-ONNX conversion can then be evaluated by comparing structural audit vectors before and after conversion.

13. Wildlife-AI Case Study Design

MegaDetector is an appropriate image-domain case study because its purpose is automated detection of animals, people, and vehicles in camera-trap imagery. Its associated publication is Beery, Morris, and Yang, Efficient Pipeline for Camera Trap Image Review, arXiv:1907.06772.

PyTorch-Wildlife provides a unified framework for wildlife monitoring, including detection models and species classifiers. Its published reference is Hernandez et al., Pytorch-Wildlife: A Collaborative Deep Learning Framework for Conservation, arXiv:2405.12930.

MegaDetector-Acoustic extends the analysis to bioacoustic monitoring through raw audio, Mel-spectrogram generation, neural classification, and sliding-window inference. This creates a useful cross-modal testbed spanning image detection, visual classification, and acoustic classification.

14. Experimental Protocol

Acquire the released PyTorch and ONNX model artifacts. Extract the computational graph and eligible tensors. For each tensor calculate $H_i = W_i^\dagger W_i$ and diagonalize it. Calculate effective rank, spectral entropy, condition measures, candidate symmetry transformations, symmetry leakage, feature manifolds, persistence diagrams, and the STTC metric. Compare the structural results with mAP, F1, accuracy, parameter count, FLOPs, and latency.

The key empirical test is whether STTC supplies statistically distinguishable information beyond conventional performance and cost metrics. Model hashes, graph versions, numerical tolerances, symmetry definitions, filtration parameters, and complete outputs should be published for reproducibility.

15. Relation to Exact Diagonalization and Quantum Computing

The diagonalization procedure is inspired by exact diagonalization in quantum many-body computation. A Hamiltonian H is decomposed into eigenstates and, when symmetries exist, invariant sectors. The neural construction analogously diagonalizes $W^\dagger W$ and decomposes tensor spaces into symmetry sectors. The analogy is mathematical rather than physical: the neural network is not asserted to be a quantum Hamiltonian.

This is relevant to Lie-algebraic quantum computing because representation theory, generators, invariant subspaces, and symmetry reduction are common mathematical structures. QuSpin provides a practical example of symmetry-resolved exact diagonalization and user-defined basis transformations.

16. Discussion

The framework changes the question from which model performs best to which model performs well with the most structurally controlled computation. A model may have excellent predictive performance while exhibiting concentrated spectra, large condition numbers, significant symmetry leakage, unstable internal representations, or high topological complexity. Conversely, a compressed model may retain comparable performance while reducing structural complexity.

Accordingly, model selection can be formulated as a multi-objective problem over performance, computational cost, and structural complexity. The STTC metric is not intended to replace accuracy or mAP, but to add a structural axis to model oversight.

17. Limitations

Neural networks do not automatically possess an exact Lie-group symmetry; the framework measures candidate symmetries. Topological quantities depend on feature-space construction, metric, filtration, and sampling. Effective rank is representation-dependent. The finite-convergence theorem requires a discrete descent condition. The weighting parameters of the composite metric require application-specific calibration. Empirical numerical values must be obtained from actual released model artifacts.

18. Conclusion

We introduced a symmetry-resolved tensor-topological framework for structural oversight of neural wildlife models, combining spectral diagonalization, symmetry decomposition, Lie-algebraic residuals, and persistent topology. The mathematical construction establishes bounded effective spectral rank and symmetry leakage, while conditional monotonicity and finite-convergence results characterize structure-preserving reduction procedures.

The central experimental question is whether neural models with comparable predictive performance can be distinguished by symmetry-resolved spectral and topological complexity. If so, tensor diagonalization and topological descriptors could provide a principled additional layer of computational oversight for scientific AI and edge-deployed biodiversity systems.

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Reproducibility and status

This manuscript is a methodological preprint draft. It does not report fabricated experimental measurements. Numerical spectra, effective ranks, symmetry leakage values, persistence diagrams, and convergence curves should be generated from the actual released model artifacts before submission as an empirical paper. This distinction is intentional and should be retained for research integrity.